An internship in

Data analytics with tableau

By smartinternz

Project name:- *Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau*

Project id:- *LTVIP2026TMIDS83621*

Date:- 20-02-2026

1. INTRODUCTION

1.1 Project Overview:-

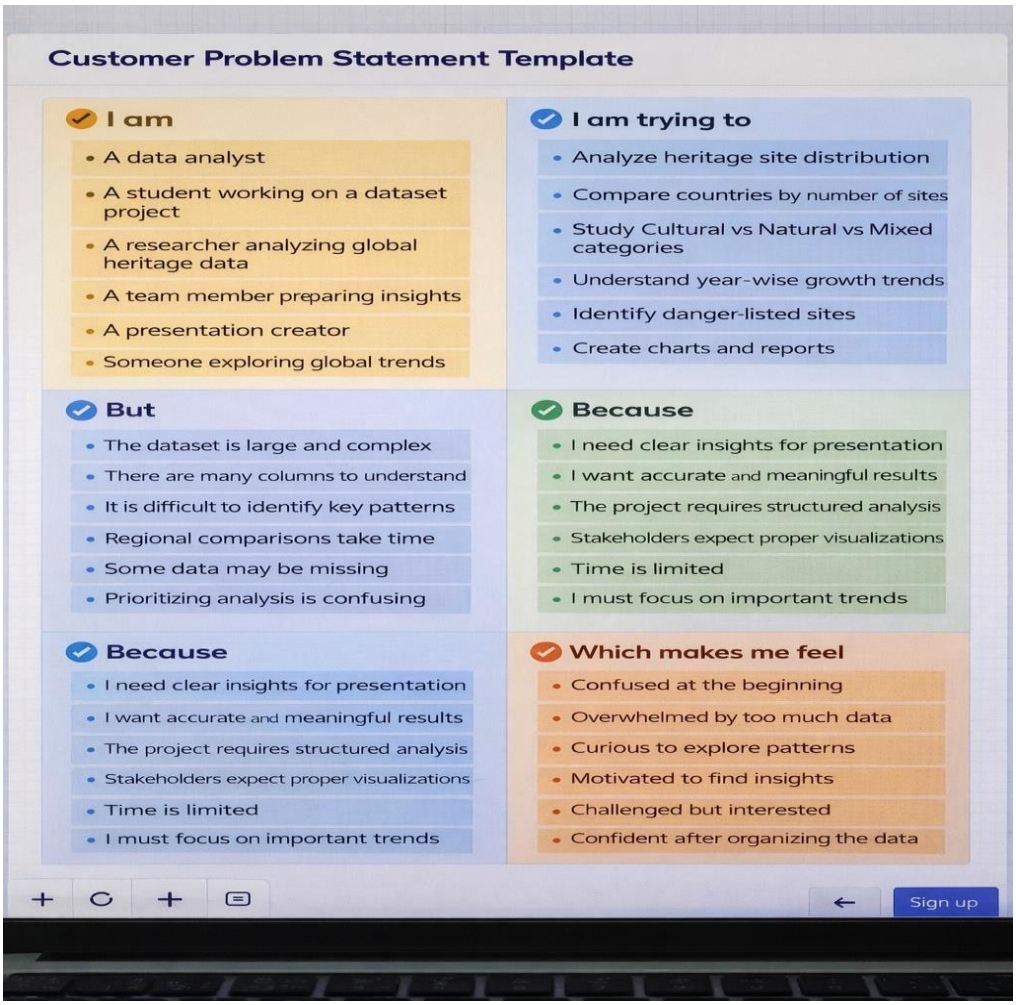
- The **Heritage Management System** is a web-based application developed to manage heritage site information and ticket bookings digitally. The main goal of the project is to reduce manual work, avoid long queues, and provide easy access to heritage site details.
- This system allows users to register, log in, view heritage site information, check ticket availability, and book tickets online. Users can also make secure online payments and receive booking confirmation.
- The admin can manage heritage site details, ticket prices, visitor records, and view reports through an admin dashboard.
- The project improves efficiency, saves time, and enhances visitor experience by providing a simple and user-friendly digital platform.

1.2 Purpose:-

- The purpose of the **Heritage Management System** is to provide a digital platform for managing heritage site information and ticket bookings.
- It helps to reduce manual work, avoid long waiting lines, and make the booking process easy and fast. The system allows users to register, view heritage site details, book tickets online, and make secure payments.
- It also helps the admin to manage site information, bookings, and visitor records efficiently.
- Overall, the purpose is to improve visitor experience and make heritage management more organized and efficient.

2. IDEATION PHASE

2.1 Problem Statement:-



Customer Problem Statement Template

I am
- A data analyst
- A student working on a dataset project
- A researcher analyzing global heritage data
- A team member preparing insights
- A presentation creator
- Someone exploring global trends

I am trying to
- Analyze heritage site distribution
- Compare countries by number of sites
- Study Cultural vs Natural vs Mixed categories
- Understand year-wise growth trends
- Identify danger-listed sites
- Create charts and reports

But
- The dataset is large and complex
- There are many columns to understand
- It is difficult to identify key patterns
- Regional comparisons take time
- Some data may be missing
- Prioritizing analysis is confusing

Because
- I need clear insights for presentation
- I want accurate and meaningful results
- The project requires structured analysis
- Stakeholders expect proper visualizations
- Time is limited
- I must focus on important trends

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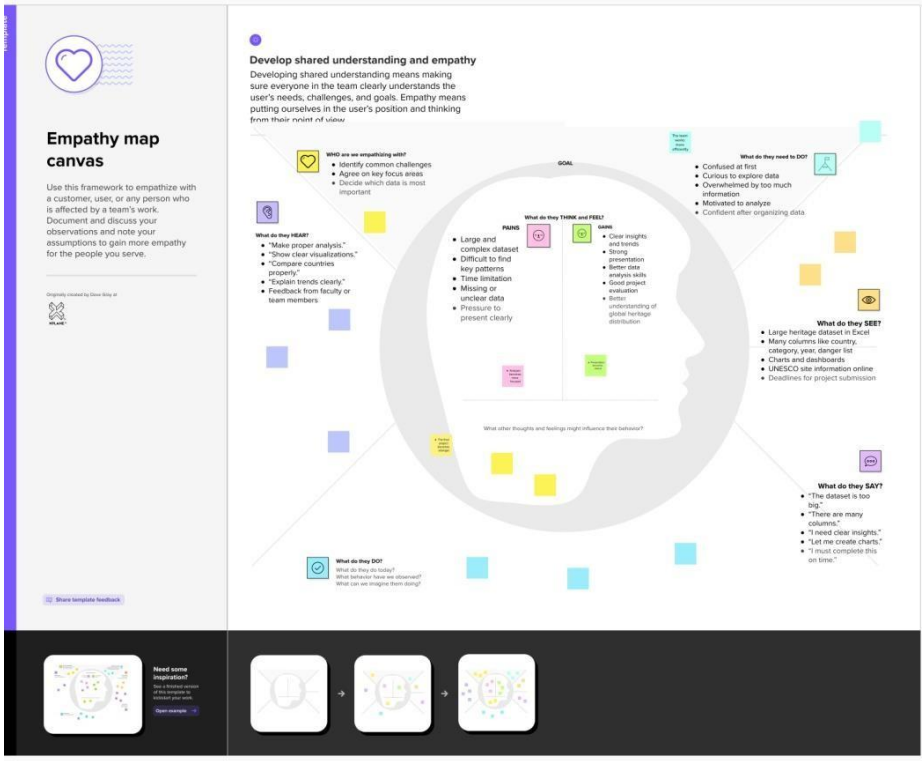
- I want accurate and meaningful results
- The project requires structured analysis
- Stakeholders expect proper visualizations
- Time is limited
- I must focus on important trends

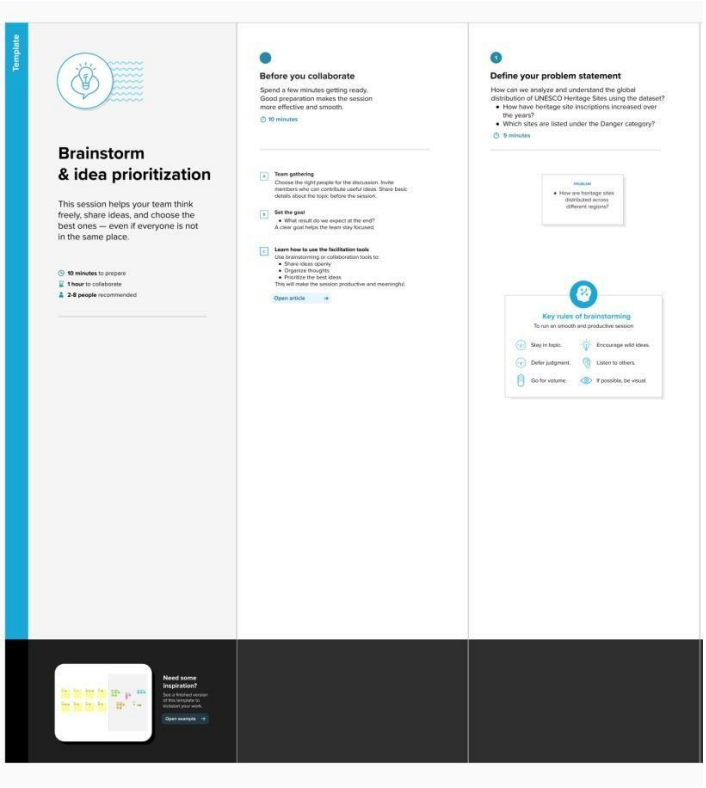
Which makes me feel
- Confused at the beginning
- Overwhelmed by too much data
- Curious to explore patterns
- Motivated to find insights
- Challenged but interested
- Confident after organizing the data

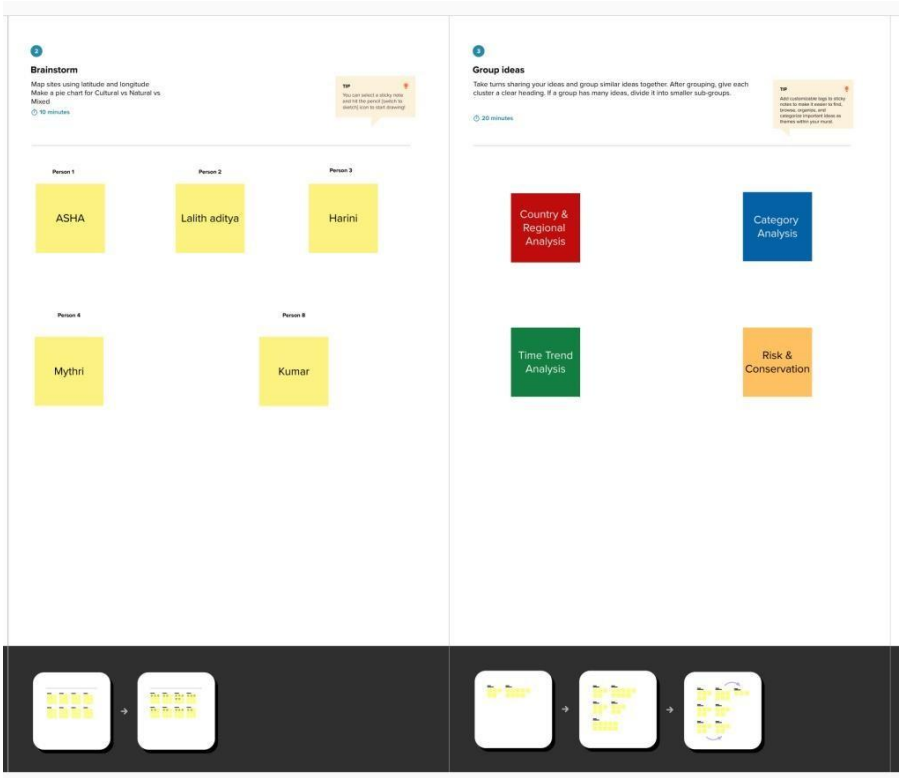
Sign up

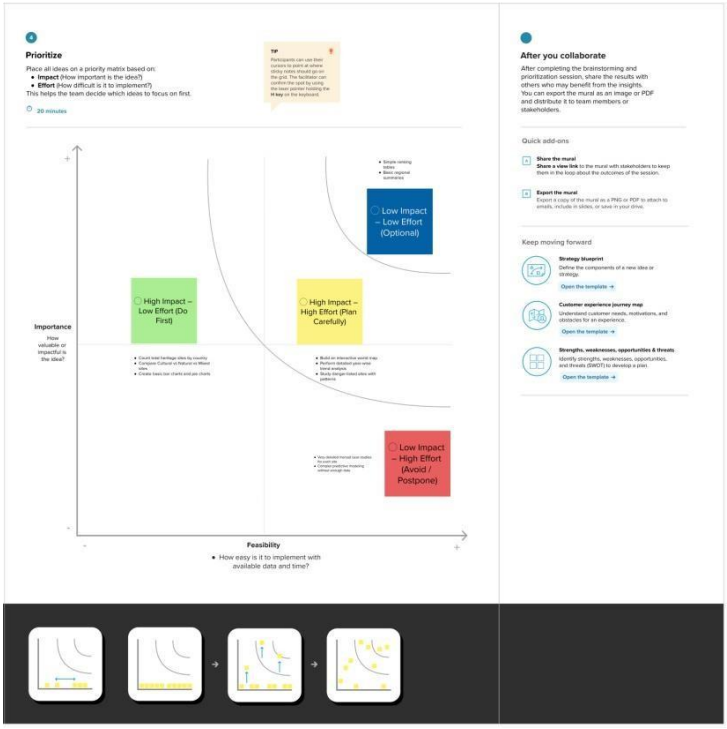
2.2 Empathy Map Canvas:



2.3 Brainstorming

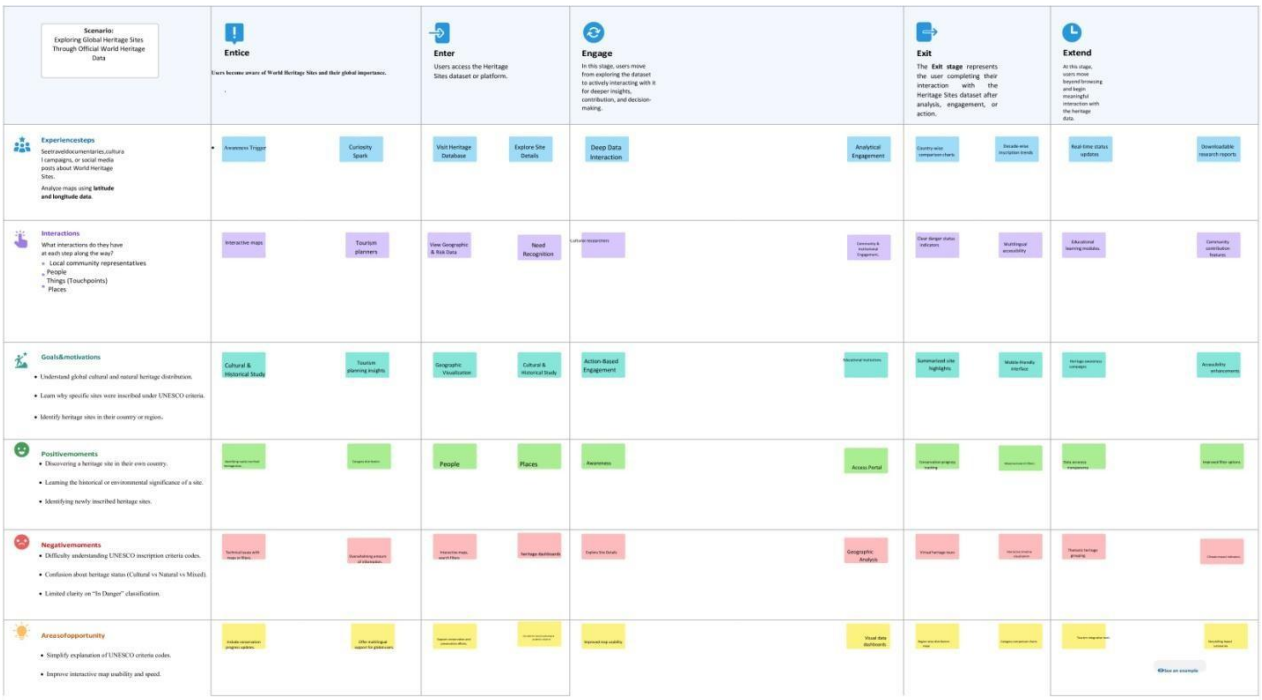



2
Brainstorm
Map sites using latitude and longitude
Make a pie chart for Cultural vs Natural vs Mixed
10 minutes
TIP
You can select a sticky note and hit the pencil (switch to draw) tool to start drawing!
Person 1
ASHA
Person 2
Lalith aditya
Person 3
Harini
Person 4
Mythri
Person 8
Kumar
3
Group ideas
Take turns sharing your ideas and group similar ideas together. After grouping, give each cluster a clear heading. If a group has many ideas, divide it into smaller sub-groups.
20 minutes
TIP
Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.
Country & Regional Analysis
Category Analysis
Time Trend Analysis
Risk & Conservation


4
Prioritize
Place all ideas on a priority matrix based on:
Impact (How important is the idea?)
Effort (How difficult is it to implement?)
This helps the team decide which ideas to focus on first.
20 minutes
TIP
Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the H key on the keyboard.
Simple ranking tables
Basic regional summaries
Low Impact – Low Effort (Optional)
High Impact – Low Effort (Do First)
High Impact – High Effort (Plan Carefully)
Low Impact – High Effort (Avoid / Postpone)
Importance
How valuable or impactful is the idea?
Count total heritage sites by country
Compare Cultural vs Natural vs Mixed sites
Create basic bar charts and pie charts
Build an interactive world map
Perform detailed year-wise trend analysis
Create region-specific sites with patterns
Very detailed thematic case studies (to avoid site
Complex pattern/time mapping without enough data
Feasibility
How easy is it to implement with available data and time?
After you collaborate
After completing the brainstorming and prioritization session, share the results with others who may benefit from the insights. You can export the mural as an image or PDF and distribute it to team members or stakeholders.
Quick add-ons
Share the mural
Share a view link to the mural with stakeholders to keep them in the loop about the outcomes of the session.
Export the mural
Export a copy of the mural as a PNG or PDF to attach to emails, include in slides, or save in your drive.
Keep moving forward
Strategy blueprint
Define the components of a new idea or strategy.
Open the template →
Customer experience journey map
Understand customer needs, motivations, and obstacles for an experience.
Open the template →
Strengths, weaknesses, opportunities & threats
Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.
Open the template →

3. REQUIREMENT ANALYSIS

Customer Journey map:-



Solution Requirement:-

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	20 february 2026
Team ID	LTVIP2026TMIDS57496
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Form • Registration through Gmail • Registration through Linked IN
FR-2	User Confirmation	• Confirmation via Email • Confirmation via OTP
FR-3	User Login	• Login using registered Email and Password • Login using Gmail account • Forgot password functionality • Reset password via email link
FR-4	Heritage Site Management	• Add new heritage site (Admin) • Update heritage site details • Delete heritage site • Upload heritage images
FR-5	Ticket Booking	• Select heritage site • Choose date and number of tickets • Calculate ticket price

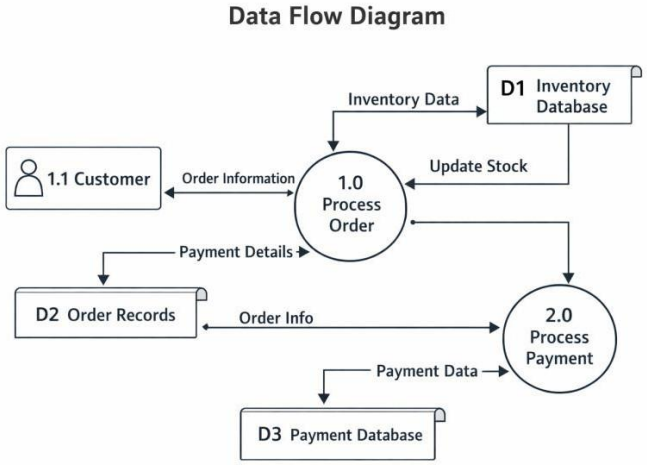
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

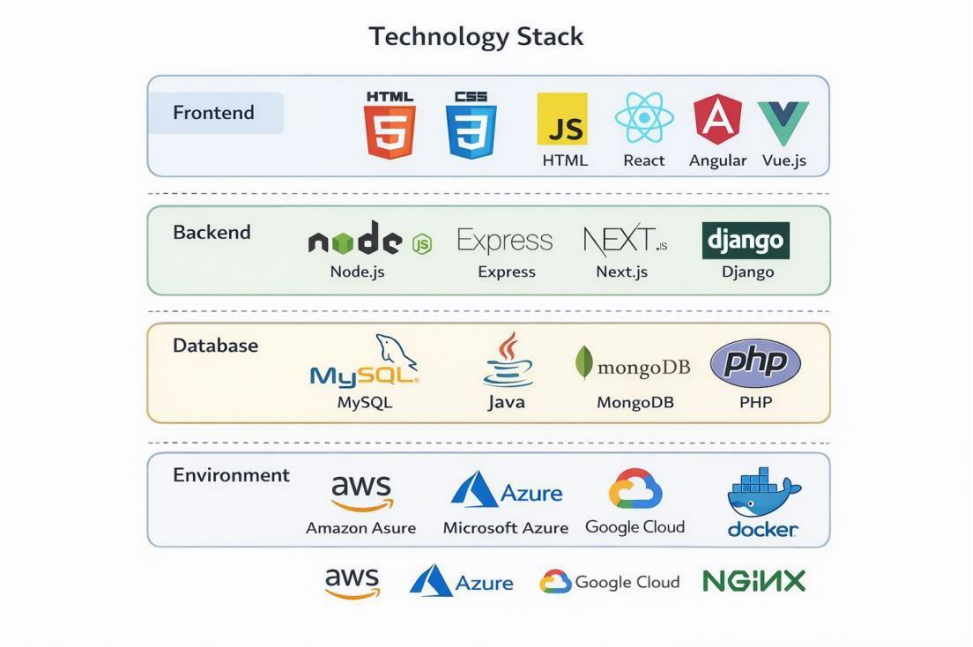
FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	• Navigation between pages (Home, Login, Dashboard, Booking, Feedback) shall be easy and intuitive. • The system shall support both web and mobile devices (responsive design). • Error messages and validation messages shall be clearly displayed to users. • Users shall be able to complete ticket booking within minimal steps.

NFR-2	**Security**	• The system shall implement secure user authentication (email/password, OTP verification). • Passwords shall be stored in encrypted format in the database. • The system shall use HTTPS protocol for secure communication.
NFR-3	**Reliability**	• The system shall operate without failure during peak usage hours. • Booking and payment transactions shall be processed accurately without data loss. • The system shall maintain transaction logs for recovery purposes. • Backup mechanisms shall be implemented to prevent data loss.
NFR-4	**Performance**	• The system shall load pages within 3–5 seconds under normal conditions. • The system shall support multiple concurrent users (e.g., 500+ users simultaneously). • Search results for heritage sites shall be displayed quickly.
NFR-5	**Availability**	• The system shall be available 24/7 for users. • Scheduled maintenance shall be informed to users in advance.
NFR-6	**Scalability**	• The system shall support increasing number of users without performance degradation. • The system shall be scalable vertically and horizontally as demand increases.

Data Flow Diagram

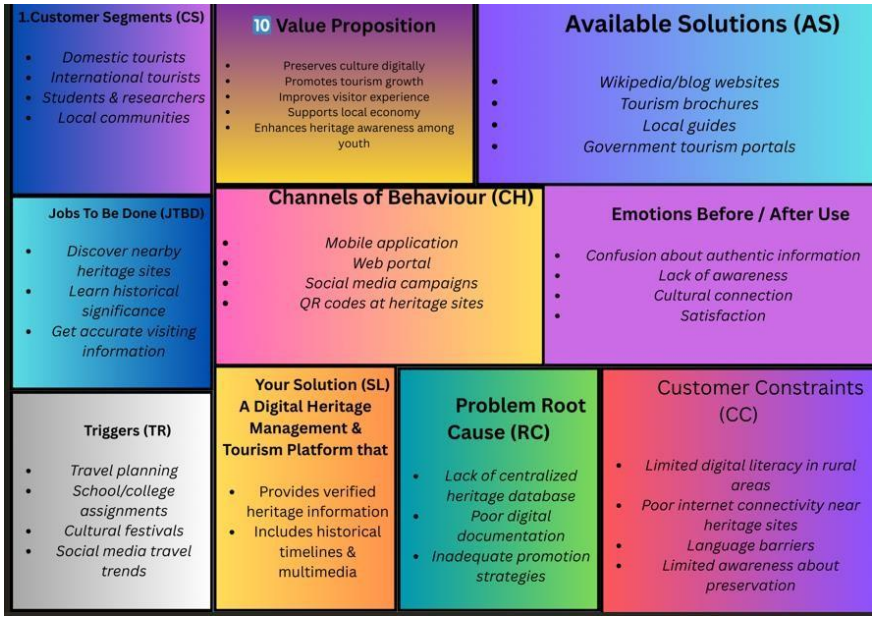


Technology Stack



4. PROJECT DESIGN

Problem Solution Fit



Proposed Solution:-

Project Design Phase-I
Proposed Solution Template

Date	20 february 2026
Team ID	LTVIP2026TMIDS57496
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many heritage sites still use manual systems for ticket booking and information. Visitors have to stand in long queues and do not get proper details about ticket prices, timings, or availability. There is no proper digital management system.
2.	Idea / Solution description	The proposed solution is an online Heritage Management System. Users can register, view heritage sites, book tickets, make online payments, and give feedback. Admin can manage site details and bookings easily.
3.	Novelty / Uniqueness	The system provides complete digital management of heritage sites in one platform. It combines online booking, secure payment, and admin management in a simple system.
4.	Social Impact / Customer Satisfaction	The system reduces waiting time, improves visitor experience, and makes heritage information easily accessible. It increases customer satisfaction through quick booking and easy access to information.
5.	Business Model (Revenue Model)	Revenue can be generated through online ticket booking charges, service fees, advertisements, and partnerships with tourism departments.
6.	Scalability of the Solution	The system can support more users and more heritage sites in the future. It can be expanded to multiple cities or countries using cloud technology.

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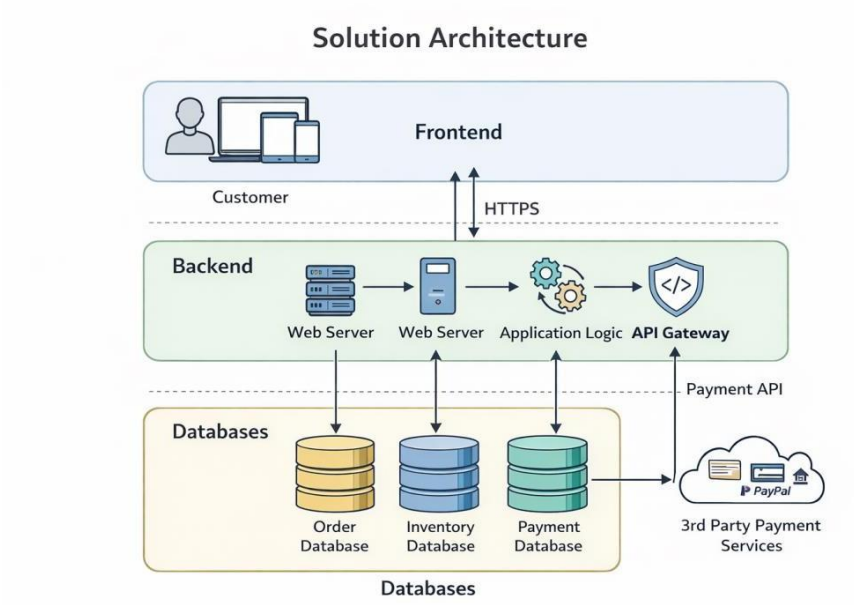
PROJECT PLANNING & SCHEDULING

Solution Architecture:-



Project Planning:-

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Storypoints)

Date	20 february 2026
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Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	ASHA
Sprint-1		USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	Lalith aditya
Sprint-2		USN-3	As a user, I can register for the application through Facebook	2	Low	Kumar
Sprint-1		USN-4	As a user, I can register for the application through Gmail	2	Medium	Mythri
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	Harini
	Dashboard					

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	10 Days	01 Dec 2025	10 Dec 2025	20	10 Dec 2025
Sprint-2	20	10 Days	15 Dec 2025	24 Dec 2025	20	24 Dec 2025
Sprint-3	20	10 Days	05 Jan 2026	14 Jan 2026	20	14 Jan 2026
Sprint-4	20	10 Days	20 Jan 2026	29 Jan 2026	20	29 Jan 2026
Sprint 5	20	10 Days	05 Feb 2026	14 Feb 2026	20	14 Feb 2026
Sprint 6	20	6 Days	15 Feb 2026	20 Feb 2026	20	20 Feb 2026

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing:-

Project Development Phase
Performance Test

Date	20 february 2026
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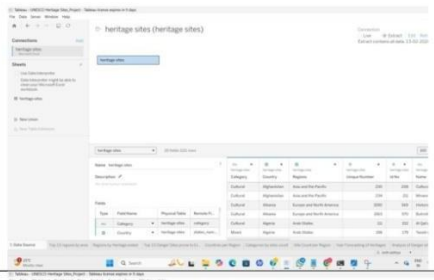
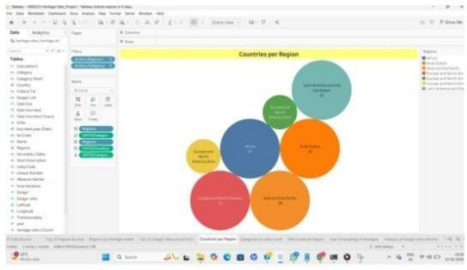
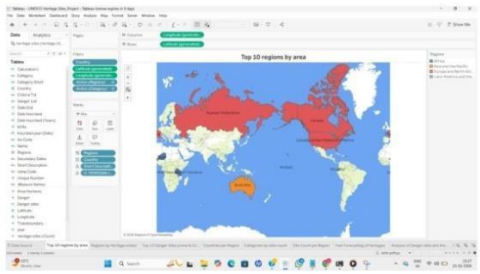
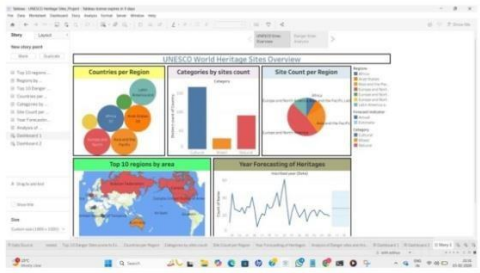
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

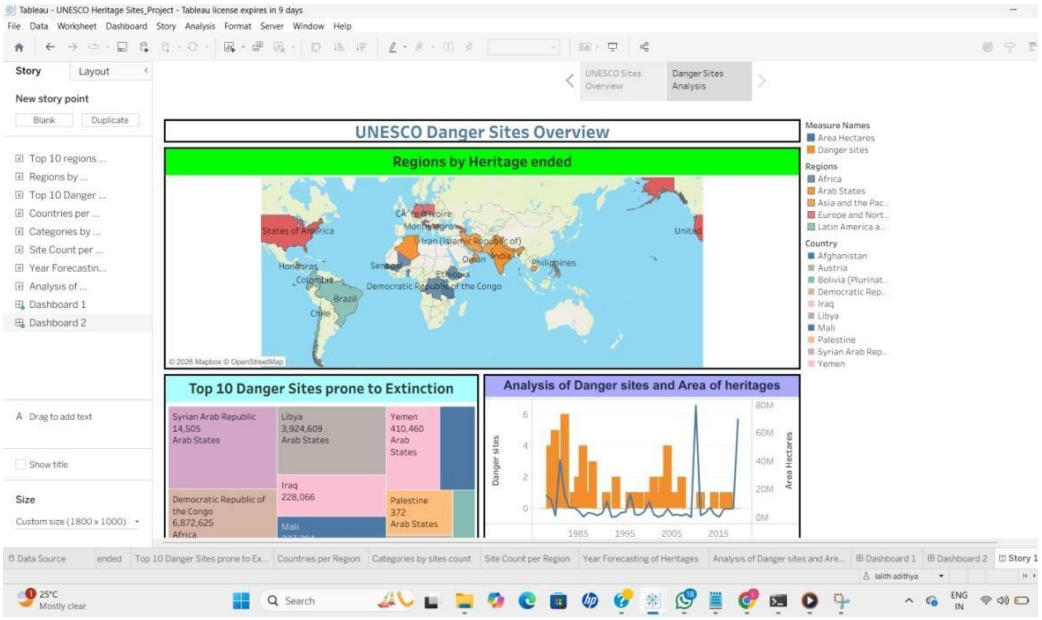
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Preprocessing	

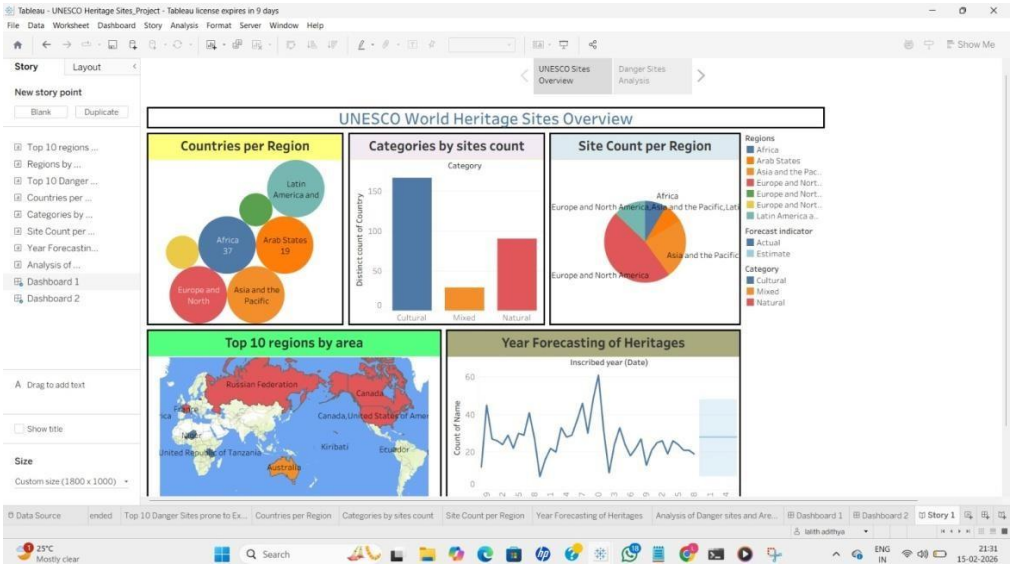
3.	Utilization of Filters	
4.	Calculation fields Used	
5.	Dashboard design	

6	Story Design	

RESULTS

Output Screenshots:-





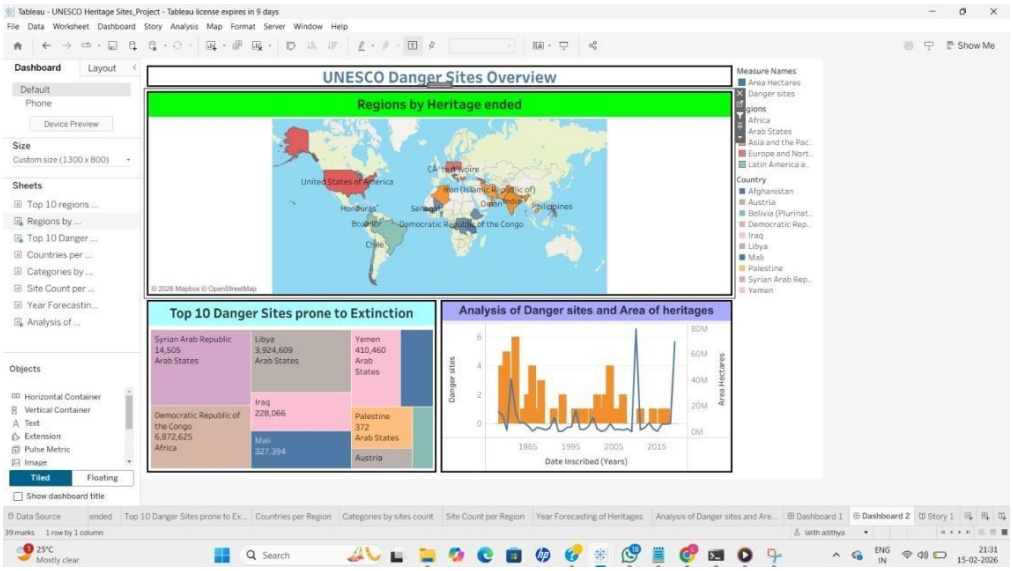


Tableau - UNESCO Heritage Sites_Project - Tableau license expires in 9 days
UNESCO Danger Sites Overview
Regions by Heritage ended
Top 10 Danger Sites prone to Extinction
Analysis of Danger sites and Area of heritages
Date Inscribed (Years)
Measure Names
Area Hectares
Danger sites
Country

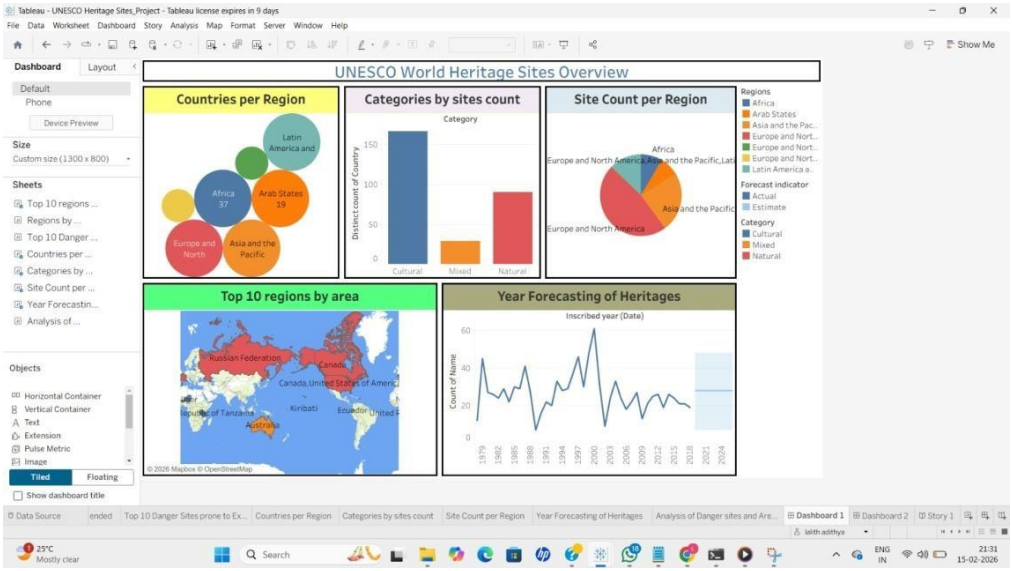


Tableau - UNESCO Heritage Sites_Project - Tableau license expires in 9 days
UNESCO World Heritage Sites Overview
Countries per Region
Categories by sites count
Site Count per Region
Top 10 regions by area
Year Forecasting of Heritages
Inscribed year (Date)
Regions
Forecast indicator
Category

ADVANTAGES & DISADVANTAGES

ADVANTAGES:-

1. **Better Planning** – Work is divided into small sprints, so it is easy to manage.
2. **Faster Delivery** – Features are completed and released quickly.
3. **Easy Tracking** – Progress can be tracked using story points and velocity.
4. **Improved Teamwork** – Team members work together regularly.
5. **Flexibility** – Changes can be made in the next sprint easily.
6. **Higher Customer Satisfaction** – Regular updates and improvements increase satisfaction.

DISADVANTAGES:-

1. **Requires Skilled Team** – Team members must understand Agile concepts.
2. **Frequent Meetings** – Daily meetings may take extra time.
3. **Scope Changes** – Too many changes can delay the project.
4. **Difficult for Large Projects** – Managing many teams can be complex.
5. **Unclear Final Cost** – Total cost may not be fixed at the beginning.

CONCLUSION

The Heritage Management System developed using Agile methodology helps in managing heritage site bookings and information efficiently.

By dividing the work into sprints, the project was completed in a planned and organized manner. The use of story points and velocity helped in tracking progress and improving team performance.

This system reduces manual work, saves time, and improves user experience through online booking and easy access to information. Overall, the project provides a simple, scalable, and effective digital solution for heritage site management.

FUTURE SCOPE

- **Mobile Application** – The system can be developed as an Android/iOS mobile app for easy access.

- **Online Payment Integration** – Add more secure payment options like UPI, credit/debit cards, and net banking.

- **Multi-Language Support** – Provide information in different languages for foreign tourists.

- **Live Ticket Availability** – Show real-time ticket availability and booking status.

- **Virtual Tour Feature** – Add 360° virtual tours of heritage sites.

- **AI Chat bot Support** – Provide instant help and guidance to users through chatbot.

- **Analytics Dashboard** – Admin can view reports on visitor count, revenue, and peak times.

- **Integration with Tourism Department** – Connect with government tourism portals for better management.

APPENDIX

- **Data set link:-** Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

https://www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites/data?select=whc-sites-2019.csv

- **Tableau public demo link:-**

  https://public.tableau.com/app/profile/lalith.adithya/viz/UNESCOHeritageSites_Project/Dashboard1

  https://public.tableau.com/app/profile/lalith.adithya/viz/UNESCOHeritageSites_Dashboard2/Dashboard2

  https://public.tableau.com/app/profile/lalith.adithya/viz/UNESCO_Project/Story1

- **Video Demo Link:-**

  https://drive.google.com/file/d/1YwOK_hLvrUoTWeDr_o3o79eNYwzzlOt2/view?usp=drivesdk